

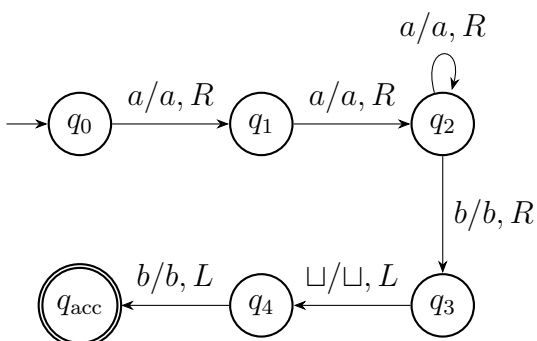
Quiz 2 Solutions — April 28, 2026

CPSC 406 — Algorithm Analysis, Section 1

Chapman University · Spring 2026

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1. (2 points) (a) Consider the following deterministic Turing machine over input alphabet $\Sigma = \{a, b\}$. The blank symbol is $\sqcup$. All omitted transitions go to a rejecting halt state.



Which of the following words is accepted by the machine? Tick exactly one.

abb *aab* *aabb* All of the above None of the above

Solution. The accepted word is *aab*.

Starting on *aab*, the machine reads the first *a* in q_0 , the second *a* in q_1 , the *b* in q_2 , moves right to the blank in q_3 , then moves left over the final *b* and accepts.

The other choices are wrong as follows. The word *abb* rejects immediately after the first *a*, because state q_1 has no shown transition on *b*. The word *aabb* reaches q_3 on the first *b*, but then sees another *b* instead of the blank, so it hits an omitted rejecting transition. Therefore neither “All of the above” nor “None of the above” is correct.

- (b) Let

$$f(n) = 7n^2 \log n + 3n + 10.$$

Which of the following Big-O classes contain $f(n)$? Tick exactly one.

$\mathcal{O}(n \log n)$ $\mathcal{O}(n^2)$ $\mathcal{O}(\exp(n))$ All of the above None of the above

Solution. The correct choice is $\mathcal{O}(\exp(n))$.

The leading term is $7n^2 \log n$, so $f(n) \notin \mathcal{O}(n \log n)$ and $f(n) \notin \mathcal{O}(n^2)$. Since every polynomial-logarithmic function is eventually bounded by an exponential function, $f(n) \in \mathcal{O}(\exp(n))$. Thus “All of the above” is wrong because the first two classes are too small, and “None of the above” is wrong because $\mathcal{O}(\exp(n))$ is correct.

2. (2 points) Show that the following formula, given in CNF, is unsatisfiable using resolution:

$$\varphi = (a \vee b \vee c) \wedge (\neg a \vee d) \wedge (\neg b \vee d) \wedge (\neg c \vee d) \wedge (\neg d \vee e \vee f) \wedge a \wedge \neg e \wedge \neg f.$$

Write the clauses and derive the empty clause $\square$.

Solution.

Let the clauses be

$$\begin{array}{ll} C_1 = a \vee b \vee c, & C_2 = \neg a \vee d, \\ C_3 = \neg b \vee d, & C_4 = \neg c \vee d, \\ C_5 = \neg d \vee e \vee f, & C_6 = a, \\ C_7 = \neg e, & C_8 = \neg f. \end{array}$$

A resolution refutation is:

$$\begin{array}{ll} C_2, C_6 \vdash d, & \text{resolving on } a, \\ C_5, d \vdash e \vee f, & \text{resolving on } d, \\ e \vee f, C_7 \vdash f, & \text{resolving on } e, \\ f, C_8 \vdash \square, & \text{resolving on } f. \end{array}$$

Thus the formula is unsatisfiable.

Common resolution mistakes. Each resolution step should use exactly two clauses and resolve on exactly one variable. Do not cancel two variables at once, do not drop unrelated literals, and do not resolve two literals with the same sign. Also, every derived clause should come from earlier clauses or original clauses, and the refutation is complete only when the empty clause $\square$ is derived.